

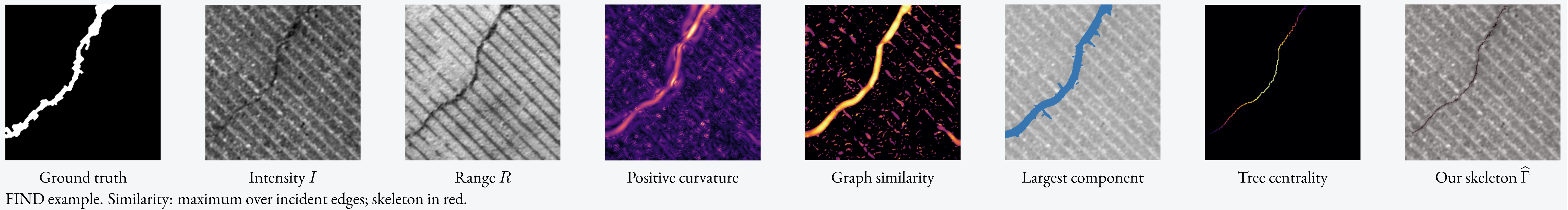
Multi-Modal, Training-Free Crack Extraction via Generalized Frangi Graph

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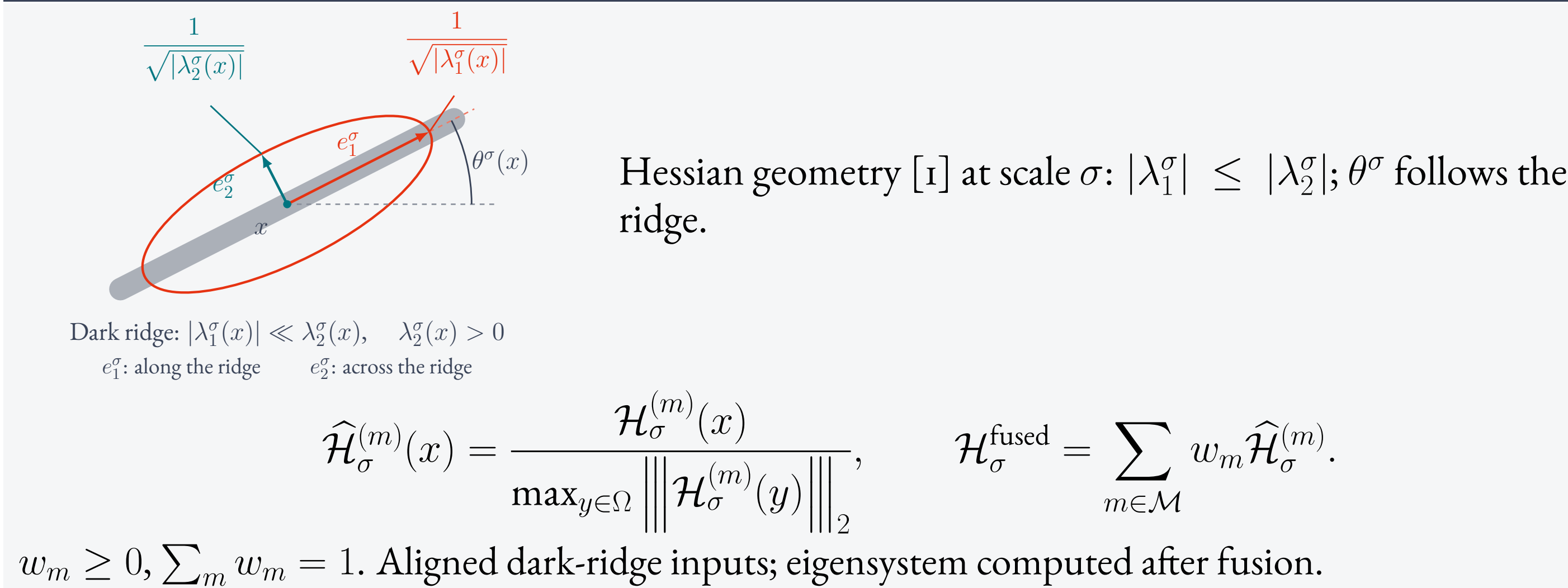
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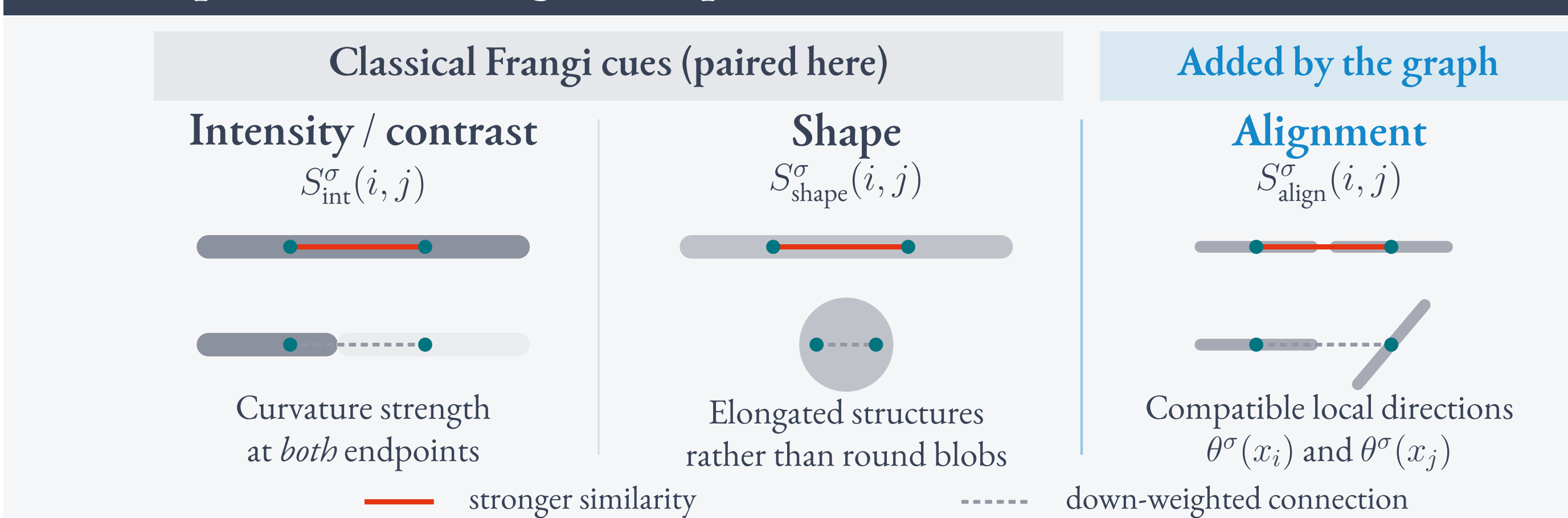
From multimodal images to a crack skeleton



1. Fuse the modalities



2. From pixel-wise Frangi [1] to pairwise links



Classical Frangi cues + neighbor alignment. For pixels with $0 < \rho_{ij} = \|x_i - x_j\|_2 \leq R$:

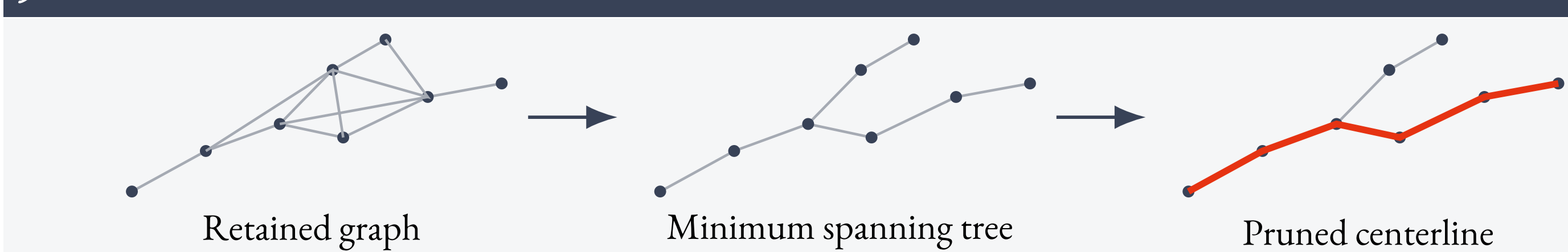
$$C_\sigma(x) = \max\{\lambda_2^\sigma(x), 0\}, \quad R_B^\sigma(x) = \frac{|\lambda_1^\sigma(x)|}{|\lambda_2^\sigma(x)|}.$$

$$S_{ij}^{(0)} = \max_{\sigma \in \Sigma} S_{\text{int}}^\sigma(i, j) S_{\text{shape}}^\sigma(i, j) S_{\text{align}}^\sigma(i, j)$$

$$d_{ij} = \min\{1, \rho_{ij}(1 - S_{ij}^{(0)})\}, \quad S_{ij} = 1 - d_{ij}.$$

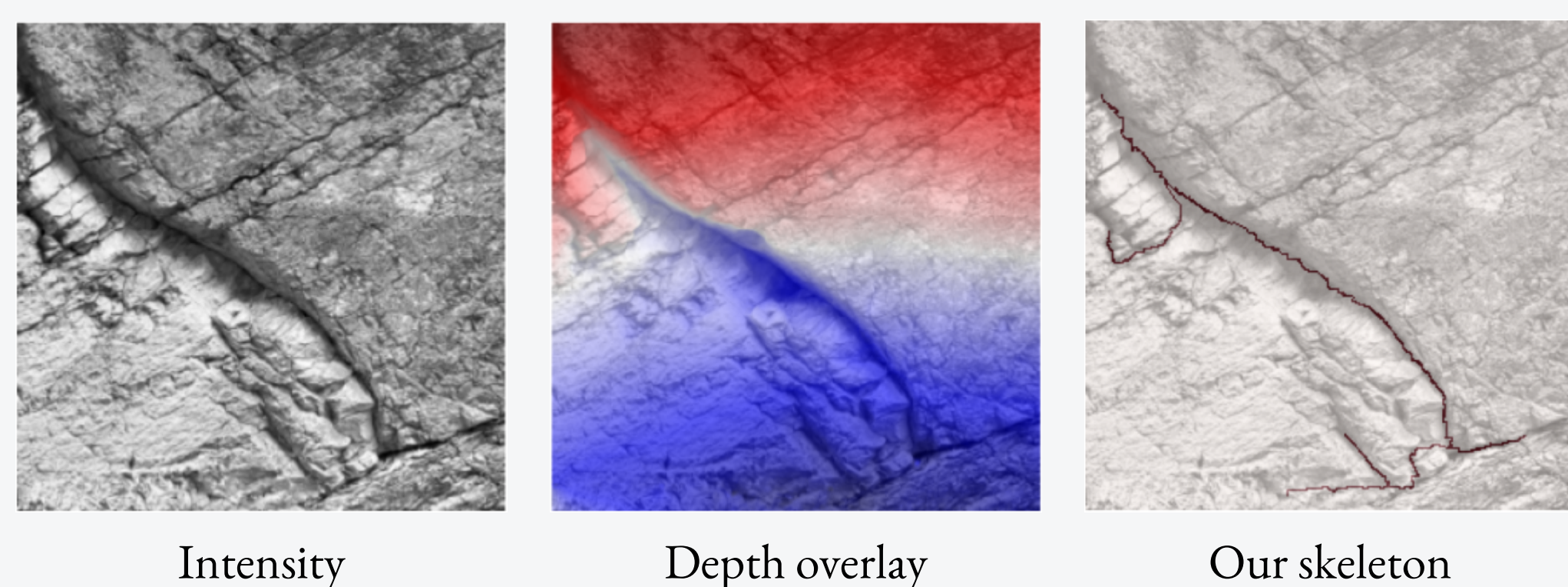
ρ_{ij} penalizes long links; θ and $\theta + \pi$ denote the same axis.

3. Extract the centerline



Per component: minimum spanning tree with costs d_{ij} , then similarity-weighted centrality. FIND settings: $\Sigma = \{1, 3, 5, 7\}$, $R = 3$, retained fractions $\tau_E = \tau_V = 0.25$.

Multimodality: Palais des Papes (France)



Fuse visible intensity and photogrammetric depth.

4. FIND: accuracy and noise

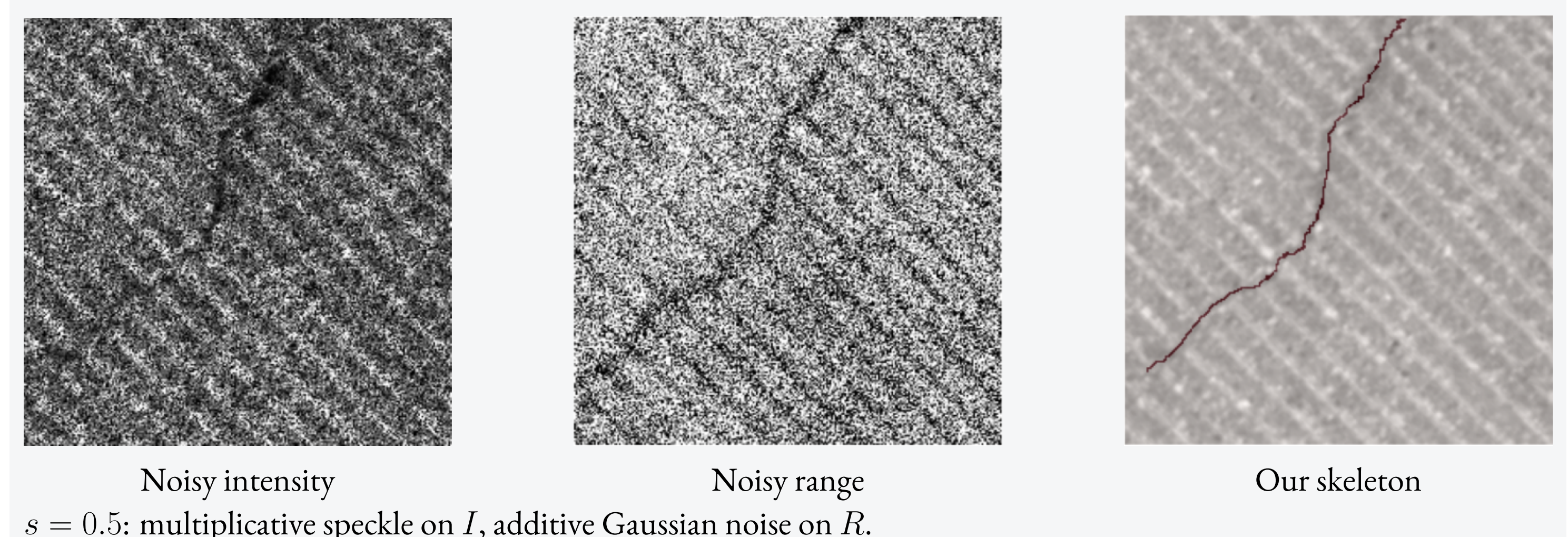
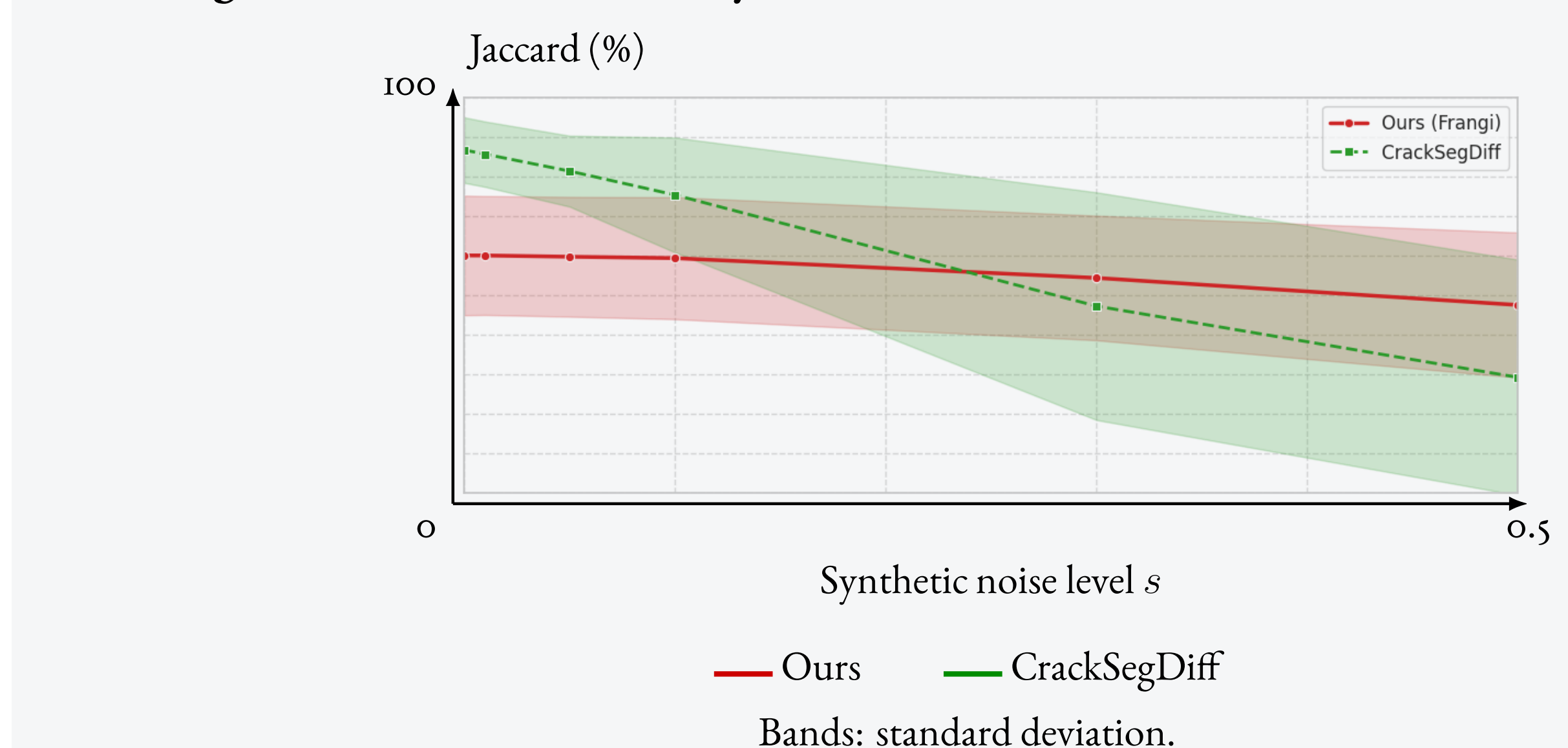
Clean FIND: the supervised model performs best.

Method ($I + R$)	Jaccard $\uparrow$	Tversky $\uparrow$	$W \downarrow$
Ours, no training	60%	68%	18 px
CrackSegDiff [3]	87%	90%	5 px

FIND [2], 256×256 . Overlap: 3 px dilation of both skeletons; Tversky: $\alpha = 1, \beta = 0.5$. W : skeleton Wasserstein distance.

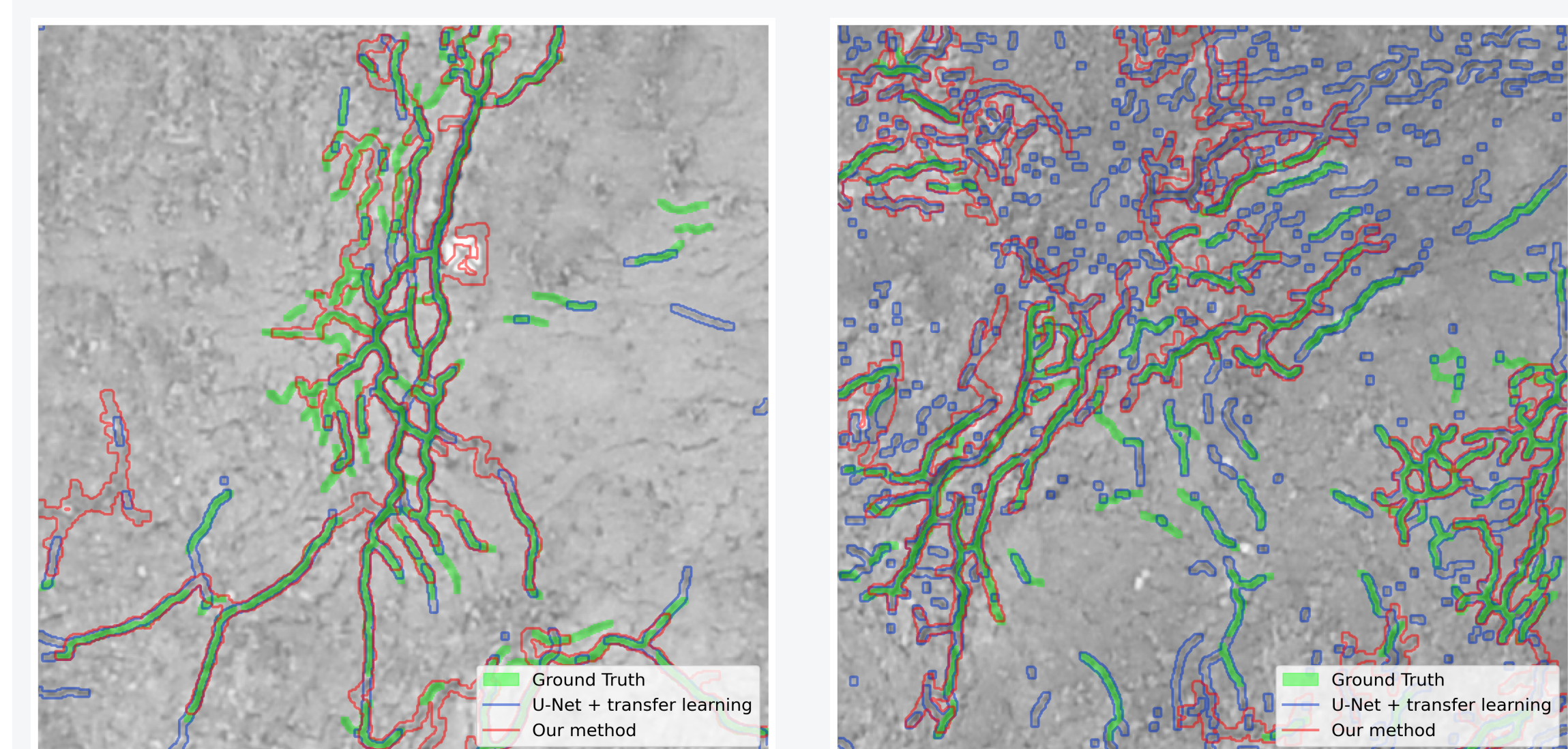
Fusion improves Jaccard: I : 41%; $I + R$: 60%; $I + R + F$: 63%. Equal weights; F is filtered range.

Slower degradation under the tested synthetic noise.



5. Geological domain shift [5]

Vaches Noires. Two Cerema UAV patches (512×512).



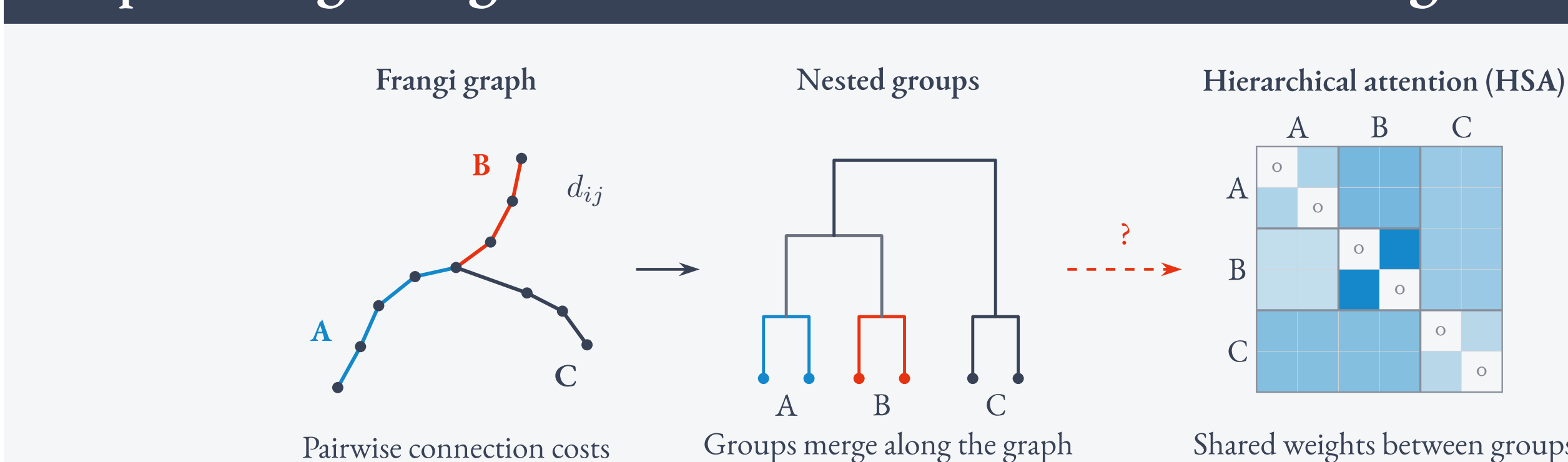
Test1: 47% / 50% Test2: 29% / 34%

— Ours — U-Net + transfer learning — Ground truth

U-Net: **about two days of expert annotation**; ours: none.

Jaccard:
ours / U-Net.
Transfer strategy [7];
local U-Net trained on
a separate image.

Perspective: guiding frozen SAM 2 / CrackSAM with a Frangi hierarchy

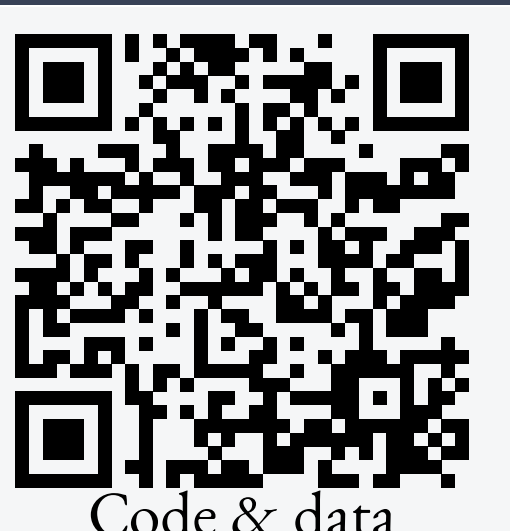


HSA [8] structures attention using an external hierarchy, without retraining in its RoBERTa experiments. Apply it to **frozen SAM 2** [6] / **CrackSAM** [4] using **Frangi groups**: segments $\rightarrow$ branches $\rightarrow$ network.

References

- [1] Frangi et al., *Multiscale vessel enhancement filtering*, MICCAI, 1998.
- [2] Zhou et al., *FIND*, Zenodo, 2022.
- [3] Jiang et al., *CrackSegDiff*, arXiv:2410.08100, 2024.
- [4] Ge et al., *Fine-tuning vision foundation model for crack segmentation in civil infrastructures*, Constr. Build. Mater. 431, 136573, 2024.
- [5] Zhang et al., *Deep Learning for Crack Detection: A Review of Learning Paradigms, Generalizability, and Datasets*, arXiv:2508.10256, 2025.
- [6] Ravi et al., *SAM 2: Segment Anything in Images and Videos*, arXiv:2408.00714, 2024.
- [7] Shi et al., *Semi-universal geo-crack detection by machine learning*, Front. Earth Sci. 11, 1073211, 2023.
- [8] Amizadeh et al., *Hierarchical Self-Attention: Generalizing Neural Attention Mechanics to Multi-Scale Problems*, NeurIPS, 2025.

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Code & data